

Photovoltaic layout, shading and storage | Models and formulas

Technical note revised for the PV Layout application, 28 September 2026. Based on the mathematical appendix by Alexandre Littardi, 25 September 2026; revised against the current source code.

What was corrected

- Dated simulations now use their actual calendar year and 365 or 366 days, rather than a fixed 2024/365 assumption. The undated Shadow preview still uses representative year 2024.
- The earlier note only described clear-sky shading. The annual Energy/BESS simulation can use imported historical weather; a chosen weather source is recorded with the results.
- The solar-to-module relation is a simplified irradiance and linear temperature model. It does not solve a single-diode circuit or reproduce a complete NOCT/SAM thermal model.
- DC voltage drop uses the lengths of both conductor runs, A+B. Replacing A+B by 2L is valid only when the two runs have equal length.
- The original MPPT voltage interval described a spread of string voltages, not electrical compatibility. Parallel strings of unequal lengths are flagged for review.
- The updated note includes hourly direct PV use, shared BESS charge/discharge power limits, SOC constraints, export limit, curtailed energy and net grid cost.
- DC conductor sections, protective devices, cold-temperature voltage, cable thermal rating, battery expansion and grid approval still require certified equipment information and engineering review.

Reading the equations

The expressions below describe the code paths used by this version. A missing weather profile selects the clear-sky estimate. Units, direction conventions and limits must be read with each equation. The guides in Home show the same material in an interactive form.

Symbols and scope

Angles are in degrees unless specified. Latitude phi is north-positive, longitude lambda east-positive, local hour h uses the supplied UTC offset u. G is W/m², P is W, E is Wh. Calculations are estimates, not equipment certification.

$$\gamma = \frac{2\pi}{N_y} \left(n - 1 + \frac{h-12}{24} \right)$$

Fractional year: N_y is 365 or 366 for the calculation year; n is the day number. The undated preview uses 2024.

Solar position

Solar elevation is corrected for refraction, while azimuth uses geometric elevation. Shadow preview uses representative year 2024 because its controls give only day/month; dated simulations and annual Energy/BESS use the real year. The annual routine derives the UTC offset for each hourly Europe/Rome timestamp.

$$E_t = 229.18(0.000075 + 0.001868\cos \gamma - 0.032077\sin \gamma - 0.014615\cos 2\gamma - 0.040849\sin 2\gamma)$$

Equation of time in minutes (Spencer/NOAA approximation).

$$\delta = 0.006918 - 0.399912 \cos \gamma + 0.070257 \sin \gamma - 0.006758 \cos 2\gamma + 0.000907 \sin 2\gamma - 0.002697 \cos 3\gamma + 0.001480 \sin 3\gamma$$

Solar declination in radians.

$$\omega = \frac{60h + E_t + 4\lambda - 60u}{4} - 180$$

Hour angle: negative in the morning, positive in the afternoon.

$$\alpha_{geo} = \arcsin(\sin \phi \sin \delta + \cos \phi \cos \delta \cos \omega)$$

Geometric solar elevation; trigonometric arguments are converted to radians.

$$\alpha = \alpha_{geo} + \frac{1.02}{60 \tan(\alpha_{geo} + 10.3/(\alpha_{geo} + 5.11))}$$

Sæmundsson refraction for geometric elevation greater than -0.85°. The tangent argument is converted to radians.

$$\cos A = \frac{\sin \delta - \sin \alpha_{geo} \sin \phi}{\cos \alpha_{geo} \cos \phi}$$

Solar azimuth from north: arccos in the morning, 360° minus arccos in the afternoon.

Irradiance and module power

A simplified clear-sky scenario is used for the shadow study. The Energy/BESS annual simulation can instead use imported historical plane-of-array irradiance AND hourly ambient temperature for a single module orientation. The 20°C ambient assumption and constant 0.20 albedo apply to the clear-sky model only.

$$m_a = \frac{1}{\sin \alpha}, \quad G_{DNI} = 1367(0.7)^{m_a^{0.675}}$$

Clear-sky direct normal irradiance; set to zero at solar elevation at or below 0.5°.

$$c_i = \max(0, \min(1, \sin \alpha \cos \beta + \cos \alpha \sin \beta \cos(A - \gamma_p)))$$

Incidence factor for module tilt beta and module azimuth gamma_p.

$$G_{POA} = G_{DNI} c_i + 0.12 G_{DNI} \sin \alpha \frac{1 + \cos \beta}{2} + 1.12 G_{DNI} \sin \alpha 0.20 \frac{1 - \cos \beta}{2}$$

Direct + simplified sky diffuse + ground reflection, with nonnegative sine of solar elevation.

$$G_{eff} = G_{POA}(1 - f_s), \quad T_c = T_{ambient} + (T_{NOCT} - 20) \frac{G_{eff}}{800}$$

Effective irradiance after fractional shading. T_ambient is 20°C for the clear-sky calculation; it comes from the hourly weather profile when supplied.

$$P = P_{STC} \frac{G_{eff}}{1000} \max(0, 1 + \frac{\mu_p}{100}(T_c - 25))$$

Linear power estimate (muP in %/°C), not an I-V or bypass-diode calculation.

Time, shading and layout

A shadow footprint is a square at the configured obstacle position. Its projection and polygon intersection are computed separately for each roof height. Real roof clearances and structural constraints are outside the grid model.

$$E = \sum_{k=1}^M P(t_{k,mid}) \Delta t_k, \quad E_{loss} = \max(0, E_{unshaded} - E_{shaded})$$

Midpoint integration; E is Wh for P in watts and time in hours.

$$E_{string} = \sum_j E_j, \quad r_{loss} = 100 \frac{\sum_j E_{loss,j}}{\sum_j E_{unshaded,j}}$$

Aggregated module energies and relative estimated string losses. This is not an electrical current-limited string model.

$$Spread = \max_j(r_{loss,j}) - \min_j(r_{loss,j})$$

Difference between the most and least affected modules of one string; a mismatch indicator only.

$$A = \frac{1}{2} \left| \sum_{i=1}^p (x_i y_{i+1} - x_{i+1} y_i) \right|$$

Shoelace area for a simple polygon; its intersection with a panel is obtained through polygon clipping.

$$d = \frac{H_{obstacle} - H_{roof}}{\tan \alpha}, \quad f_s = \min\left(1, \frac{A_{intersection}}{A_{panel}} o\right)$$

Shadow length at positive elevation and positive height difference; o is obstacle opacity in [0,1].

$$n_{col} = \lfloor \frac{W}{w} \rfloor, \quad n_{row} = \lfloor \frac{H}{h} \rfloor$$

Panel grid before user edits; alignment uses residual width/height, and rotation acts about each panel centre.

$$o_x \in \{0, (W - n_{col}w)/2, W - n_{col}w\}, \quad o_y \in \{0, (H - n_{row}h)/2, H - n_{row}h\}$$

Left/centre/right and top/centre/bottom alignment of the residual clearance.

$$x' = c_x + (x - c_x)\cos\theta - (y - c_y)\sin\theta$$

Rotation of x around each panel centre.

$$y' = c_y + (x - c_x)\sin\theta + (y - c_y)\cos\theta$$

Rotation of y. The convex hull and polygon clipping use Andrew and Sutherland-Hodgman algorithms.

Strings, routing and voltage drop

N modules in series multiply voltage but retain module current. Parallel MPPT strings add currents; unlike a loose voltage interval, differing string voltages need an explicit mismatch check. Routes A and B are separate physical conductor paths.

$$P_{string} = \frac{NP_{STC}}{1000}, \quad V_{oc,string} = NV_{oc}, \quad V_{mpp,string} = NV_{mpp}$$

kWp and voltages at STC; Isc and Imp each equal the module current for a series string.

$$I_{MPPT} = \sum_{j=1}^k I_{string,j}, \quad P_{inv,DC} = \sum_{s \in inv} P_{string,s}$$

Parallel current and assigned DC peak power. Check maximum voltage at cold temperature separately.

$$L_{loop} = L_A + L_B, \quad \Delta V = \frac{\rho L_{loop}}{S}$$

Two conductor lengths including modeled vertical drop and reserves; rho is $\Omega \cdot \text{mm}^2/\text{m}$, S is mm^2 .

$$L_{orthogonal} = |x_2 - x_1| + |y_2 - y_1|$$

Fallback Manhattan route between two points without a drawn path; the drawn cable network uses Dijkstra shortest-path search.

$$V_{oc, cold} = NV_{oc, STC} \left[1 + \frac{\mu_{Voc}}{100} (T_{min} - 25) \right]$$

Cold-voltage estimate where the module voltage temperature coefficient and site design temperature are supplied; otherwise the check remains open.

$$S_{min} = \frac{\rho(L_A + L_B)}{U \epsilon_{target}/100}, \quad \epsilon_{actual} = 100 \frac{\rho(L_A + L_B)}{SU}$$

Equivalent to 2L only if $L_A = L_B = L$. Sizing is voltage-drop-only; no ampacity or fault study.

$$I_{AC, nom} = \frac{1000 S_{AC, kVA}}{\sqrt{3} U_{LL}}$$

Estimated three-phase line current from apparent inverter rating and entered line voltage; this is not a breaker selection.

Hourly PV, BESS and economics

For each hour the simulator serves load directly from PV, then charges storage from surplus, then exports up to the requested grid limit. Storage power limits are shared, not multiplied by the number of cabinets; SOC starts at its minimum. No grid charging or battery resale is modeled.

$$E_{direct} = \min(E_{PV}, E_{load}), \quad E_{surplus} = E_{PV} - E_{direct}$$

Direct self-consumption and potential excess PV energy.

$$E_{charge} \leq \min(E_{surplus}, P_{charge} \Delta t, E_{room}/\eta_c)$$

Charge constrained by surplus, power and remaining usable battery capacity.

$$E_{discharge} \leq \min(E_{load} - E_{direct}, P_{discharge} \Delta t, E_{usable} \eta_d)$$

Battery discharge constrained by residual load, shared power cap and stored energy.

$$E_{SOC, next} = E_{SOC} + \eta_c E_{charge} - \frac{E_{discharge}}{\eta_d}, \quad \eta_c = \eta_d = \sqrt{\eta_{roundtrip}}$$

Hourly storage transition with initial SOC at its configured minimum. The code currently uses a one-hour time step.

$$E_{grid} = E_{load} - E_{direct} - E_{discharge}, \quad E_{export} = \min(E_{surplus} - E_{charge}, P_{export} \Delta t)$$

Residual purchase and capped export. Curtailment is excess beyond that export limit.

$$C_{net} = p_{buy} E_{grid} - p_{sell} E_{export}, \quad Savings = p_{buy} E_{load} - C_{net}$$

Annual values use summed hourly energies; battery benefit includes foregone export revenue.

Traceability and limits

- Solar geometry and irradiance: mixins/shadow_energy.py; obstacle footprint, clipping and area: mixins/shadow_geometry.py; panel grid: mixins/zone_tools.py.
- String and MPPT STC values: mixins/diagram_tools.py and detailed_electrical.py; network routes and two DC terminals: mixins/cable_network.py and mixins/two_pole_cables.py; voltage-drop recommendation: mixins/notes_tools.py.
- Hourly generation: self_consumption.py; one/two BESS dispatch: battery_dispatch.py; avoided purchases, export revenue and payback: energy_economics.py.
- The roof shadow model uses a square obstacle footprint and separate zone heights, not an as-built mesh of a lattice tower. String mismatch, diodes, snow, wind, bifacial gain and all network protection settings are outside this simulation.
- Public reference: NOAA General Solar Position Calculations, <https://gml.noaa.gov/grad/solcalc/solareqns.PDF> (366-day leap-year denominator).
- Public reference: Sandia PV Performance Modeling Collaborative, NOCT Cell Temperature, <https://pvpmc.sandia.gov/modeling-guide/2-dc-module-iv/cell-temperature/noct-cell-temperature/> (the application uses a simplified ambient-temperature estimate).

Interpretation for the 516-module project

- The supplied 516-module reference represents 371.52 kWp of nominal DC modules assigned to three 125 kVA hybrid inverters and thirty-one strings. The stored annual clear-sky calculation is a scenario, not measured PV yield; historic weather and load data must be checked for matching dates and time conventions.
- One MC-L522 has two battery clusters connected to two inverter BAT ports in the available DEYE drawing. The third hybrid is dedicated to PV. An additional MC-L522 cabinet must not be represented as approved parallel DC wiring until a manufacturer-approved extension drawing is available.
- The requested 300 kW export setting is a scenario limit for the hourly dispatch. The 507 kW existing contract is a different quantity. Neither value by itself documents a granted injection authorization or a tested grid-interface protection.
- Cable paths are estimates until traced on site. A valid A+B record includes two conductor routes, roof height difference and configured reserve. Conductor sizing from DC voltage drop does not substitute for ampacity, short-circuit, thermal and protective coordination studies.
- Shadow preview and dated shadow studies accept a manually supplied UTC offset; the undated preview uses representative year 2024. Dated shadow simulations use the real calendar year. The annual simulation derives hourly offsets from Europe/Rome and its timestamp. Clear-sky module temperature assumes 20°C ambient; imported weather supplies hourly ambient temperature and plane-of-array irradiance for a single module orientation.